

H2. Inception

✓ Full solution

Time limit	3 seconds
Memory limit	256Mb
Input	standard input or input.txt
Output	standard output or output.txt

Aliens attack city χ ! The city has n buildings, each building is a point with coordinates (x_i, y_i) . Aliens can conquer any square with sides parallel to coordinate axis. They want to conquer at least k buildings. But conquering of big area is tedious and time consuming, so aliens want to minimize sidelength of the square.

Please help aliens to find the smallest square, containing at least k buildings.

Input format

First line of input contains two integers n ($2 \leq n \leq 50\,000$) — number of buildings — and k ($2 \leq k \leq n$) — minimal number of buildings aliens want to conquer. Next n lines contain coordinates of points of buildings x_i and y_i ($1 \leq x_i, y_i \leq 10\,000$) — integer numbers, separated by space.

Output format

In the first line output single integer A — minimal sidelength of the square, containing at least k points. In the second line output two integers — coordinates x and y of the bottom-left corner of the square (so, square will contain all points (x_i, y_i) that satisfy $x \leq x_i \leq x + a$ and $y \leq y_i \leq y + a$). If there are multiple solutions, output any.

Sample

Input	Output
3 3	3
1 2	1 1
2 1	
3 4	